Cross-Modal Change Attention and Damage-Aware CutMix for Multimodal Building Damage Assessment

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***Abstract*—Multimodal building damage assessment requires reliable fusion between pre-event optical imagery and post-event synthetic aperture radar (SAR) observations, while also coping with severe damage-class imbalance and cross-event distribution shifts. This paper introduces a cross-modal change attention module and a damage-aware CutMix training strategy for all-weather building damage mapping. The method is designed as a lightweight plug-in for standard segmentation backbones and is evaluated through controlled ablation studies on the BRIGHT benchmark.**

***Index Terms*—Building damage assessment, multimodal remote sensing, optical imagery, synthetic aperture radar, cross-modal attention, data augmentation, CutMix.**

I. Introduction

RAPID building damage assessment (BDA) is an important component of emergency response after earthquakes, floods, storms, wildfires, and other disasters. Building-level damage maps support search-and-rescue prioritization, loss estimation, and relief allocation, yet field surveys are often slow, dangerous, and spatially incomplete. Earth observation provides a scalable means of repeatedly observing affected areas, and optical, synthetic aperture radar (SAR), and multitemporal imagery have therefore been widely investigated for post-disaster mapping [1]–[3]. More recently, very-high-resolution (VHR) imagery and large annotated datasets have shifted BDA from handcrafted indicators toward data-driven building localization and multiclass damage recognition [4]. Deep models now support joint localization and classification, multitemporal feature interaction, and object-based semantic change detection across multiple hazards [5]–[7].

Most large-scale BDA systems nevertheless depend on paired pre- and post-event optical images. Optical imagery provides intuitive spectral, textural, and geometric evidence, but a usable post-event acquisition may be delayed by clouds, smoke, precipitation, or darkness. SAR actively illuminates the surface and can observe independently of daylight and through most weather conditions. It is therefore especially valuable immediately after a disaster, when a pre-event SAR image with matching geometry may not exist. Pairing detailed pre-event optical imagery with rapidly available post-event SAR imagery offers a practical alternative. Earlier studies demonstrated the feasibility of multimodal optical–SAR damage mapping [8], [9], while the BRIGHT benchmark establishes a globally distributed VHR setting with paired imagery and pixel-level damage labels [10].

This setting poses a challenging heterogeneous change-understanding problem. Optical values describe reflected solar radiation, whereas SAR amplitudes depend on microwave scattering, viewing geometry, surface roughness, and dielectric properties. Consequently, corresponding pixels can have very different appearances even after geometric coregistration. Direct channel concatenation gives a network access to both modalities, but does not explicitly indicate which post-event SAR responses should be compared with the pre-event optical structure. The learned representation may therefore entangle actual structural damage with cross-sensor discrepancy. Heterogeneous change-detection and optical–SAR segmentation methods reduce this gap through image translation, common feature spaces, or progressive alignment [11]–[13]; however, these approaches mainly target binary land-cover change or static building extraction rather than four-class damage assessment across diverse events.

A second difficulty is the joint effect of class imbalance and cross-event distribution shift. Background and intact buildings dominate the training pixels, whereas damaged and destroyed buildings are sparse and highly event-dependent [14]. Their signatures vary with hazard mechanism, construction type, sensor configuration, and geography, so strong aggregate performance can conceal weak recognition of the most operationally important categories. Event-disjoint evaluations further show that models can deteriorate substantially on unseen locations [15]–[17]. Generic geometric and radiometric augmentation increases appearance diversity, but it neither guarantees sufficient exposure to rare damage pixels nor forces the model to observe similar damage evidence under different disaster contexts.

These two difficulties are coupled. If a network does not explicitly account for the optical–SAR modality gap, cross-sensor differences can dominate the weak evidence of rare damage classes. Conversely, improving feature fusion alone does not ensure that the network observes enough damaged and destroyed examples from diverse disaster contexts. Existing work has mainly addressed heterogeneous fusion or class imbalance as separate problems. A practical all-weather BDA system therefore requires complementary model-level and data-level mechanisms that can be evaluated independently and attached to different backbones without redesigning the complete segmentation pipeline.

To address these limitations, this paper proposes a Cross-Modal Change Attention (CMCA) module and a Damage-Aware CutMix (DA-CutMix) training strategy. CMCA uses

pre-event optical features as spatial queries and post-event SAR features as keys and values, enabling explicit optical-to-SAR comparison at the semantic bottleneck. The attention response retrieves SAR evidence according to the pre-disaster building structure rather than treating both modalities as interchangeable channels. DA-CutMix extends CutMix [18] by sampling damage-rich donor patches from different disaster events and pasting them synchronously into the optical image, SAR image, and label map. This cross-event construction increases rare-class exposure while preserving the correspondence among the two sensor streams and dense supervision. The two components address feature fusion and data imbalance, respectively, while remaining lightweight enough to attach to multiple segmentation backbones.

The main contributions of this paper are summarized as follows.

- We propose CMCA, a lightweight optical-to-SAR attention module that replaces simple modality concatenation with explicit change-aware feature interaction at the semantic bottleneck.
- We introduce DA-CutMix, an event-aware and damage-aware augmentation strategy that selects damage-rich cross-event patches and preserves optical–SAR–label alignment during mixing.
- We conduct controlled ablations on the BRIGHT benchmark using six representative CNN-, transformer-, Siamese-, recurrent-, and Mamba-based backbones, and analyze both aggregate performance and per-event robustness.

The remainder of this paper is organized as follows. Section II reviews remote sensing BDA, change-detection architectures, and multimodal fusion. Section III presents CMCA, DA-CutMix, and the training objective. Section IV describes the experimental protocol and analyzes the overall and per-event results. Section V concludes the paper.

II. Related Work

A. Building Damage Assessment and Change Detection

Remote sensing BDA has traditionally relied on spectral, textural, geometric, and object-level differences in optical imagery, together with backscatter intensity, coherence, and polarimetric scattering in SAR observations [1]–[3]. Multitemporal PolSAR studies linked changes in double-bounce and volume scattering to structural damage [19], [20], while other approaches combined intensity, coherence, or superpixel evidence from multiple sensors [21], [22]. These methods established the value of SAR for rapid response, but their performance often depends on sensor configuration, acquisition geometry, and handcrafted descriptors.

Large annotated datasets have shifted BDA toward deep dense prediction. xBD introduced bitemporal optical imagery with building footprints and multilevel damage labels [4]. RescueNet jointly learned building localization and damage assessment [5], BDANet combined multiscale fusion, cross-directional attention, and class-focused CutMix [6], and ChangeOS formulated BDA as object-based semantic change detection [7]. Subsequent evaluations showed that models remain sensitive to unseen locations, disaster types, and intermediate damage classes [16], [17].

Related change-detection architectures provide useful mechanisms for paired-image reasoning. Convolutional Siamese networks learn feature differences directly [23], SiamCRNN introduces recurrent interaction for multisource VHR imagery [24], and Transformer models such as BIT and DamFormer capture nonlocal temporal relationships [25], [26]. More recently, ChangeMamba has adapted state-space modeling to binary change detection, semantic change detection, and BDA [27]. These developments motivate testing the proposed modules across diverse backbones, but they do not by themselves resolve the heterogeneous optical–SAR modality gap or the scarcity of severe-damage samples.

B. Multimodal Optical–SAR Fusion

Optical–SAR fusion has long been used in BDA to combine complementary observations of the same affected area. Early approaches integrated geometric and spectral cues from VHR optical imagery with intensity, coherence, or scattering information from SAR data. Joint optical–SAR analysis was applied to earthquake damage assessment [28], and superpixel-based belief fusion combined decisions derived from the two sensors [22]. With the development of deep learning, multimodal and multitemporal features have been learned jointly for building damage mapping [8]. More recently, M3ICNet introduced cross-modal and cross-resolution interaction for optical–SAR damage detection [9], while BRIGHT provided globally distributed paired data and dense annotations for learning all-weather BDA models [10].

Related remote sensing tasks offer further insight into heterogeneous feature fusion. SpaceNet 6 established a paired optical–SAR setting for all-weather building mapping [29], and progressive fusion learning combined modality-specific and shared representations for joint building segmentation [13]. For heterogeneous change detection, deep homogeneous feature fusion maps sensor-dependent observations into a common representation before change comparison [11], whereas progressive modality alignment alternates feature alignment and change-map refinement [12]. Although these methods address the optical–SAR gap, they are primarily designed for static building extraction or binary change detection rather than multiclass damage assessment.

Attention-based fusion enables the contribution of one modality to vary with image content. The query–key mechanism provides a general formulation for retrieving relevant information across feature sets [30], and change-guided attention has been used to inject change priors into multiscale remote sensing features [31]. Existing fusion networks, however, are commonly tied to a particular architecture or rely on global modality alignment. In contrast, the proposed CMCA performs directional optical-to-SAR interaction at a deep semantic stage and exposes a compact feature interface, allowing the same fusion operation to be attached to different BDA backbones.

III. Proposed Method

This section presents two complementary and independently pluggable components for multimodal building damage assess-

ment. Cross-Modal Change Attention (CMCA) is a model-level plug-in that receives paired optical and SAR feature maps at a selected encoder stage and returns a change-aware feature map with a compatible spatial resolution. It can therefore be inserted between an encoder and its original decoder or prediction head through a small output adapter. Damage-Aware CutMix (DA-CutMix) is a data-level plug-in that transforms a training triplet before it enters the network; it does not alter the backbone or the inference procedure. The first contribution makes cross-modal comparison explicit without prescribing a new decoder, while the second increases rare-damage exposure with no inference-time cost. The complete data flow is shown in Fig. 1, using a UNet-style encoder–decoder as a representative visualization; the same plug-in interface is adapted to the other evaluated backbones.

A. Problem Formulation

Following the multimodal BDA setting of BRIGHT [10], each sample contains a geometrically coregistered pre-event RGB optical image and a post-event single-band SAR amplitude image. Let H and W denote their height and width in pixels. The optical input is written as $\mathbf{X}^o \in \mathbb{R}^{3\times H\times W}$, whose three channels correspond to the red, green, and blue bands. To use the same three-channel encoder interface for both modalities, the SAR amplitude band is repeated along the channel dimension to form $\mathbf{X}^s \in \mathbb{R}^{3\times H\times W}$. The three SAR channels therefore contain identical measurements and do not represent different polarizations.

Given the optical–SAR pair, the objective is to learn a segmentation model f_θ that predicts a pixel-wise damage map

$$\hat{\mathbf{Y}} = f_\theta\left(\mathbf{X}^o, \mathbf{X}^s\right), \tag{1}$$

where θ denotes the learnable model parameters and $\hat{\mathbf{Y}} \in \{0,1,2,3\}^{H\times W}$ is the predicted label map. Labels 0–3 represent background, intact building, damaged building, and destroyed building, respectively. The ground-truth map $\mathbf{Y}$ has the same spatial dimensions as $\hat{\mathbf{Y}}$, and the model is trained for four-class semantic segmentation.

B. Overall Framework

The framework follows the input-to-output sequence below. First, a pre-event optical image, a post-event SAR image, and their damage label map are loaded on the same pixel grid. During training, the DA-CutMix plug-in may replace a damage-rich rectangular region of this triplet with a region from another disaster event. The three layers are edited together so that every inserted optical pixel, SAR pixel, and class label still refers to the same donor location. When DA-CutMix is disabled or when no valid donor patch is found, the original triplet proceeds unchanged.

Second, the optical and SAR images are processed by separate encoder streams, allowing each branch to learn modality-specific representations before cross-modal interaction. At successive encoder stages, the two streams produce paired feature maps at corresponding spatial resolutions. For encoder–decoder backbones with skip connections, shallow paired features are projected and fused before being passed to the decoder, whereas the deepest feature pair is sent to CMCA for explicit optical–SAR interaction.

Third, CMCA uses every optical bottleneck position to retrieve relevant post-event evidence from the SAR bottleneck. It then compares the retrieved SAR representation with the optical representation and returns a change-aware feature map. A lightweight adapter injects this output into the base architecture while preserving the tensor shape expected by the original decoder or prediction head. Depending on the backbone, this adapter concatenates and projects the features, replaces the original deepest fusion, or adds a projected residual. Consequently, insertion requires exposing one paired feature level and changing only its fusion operation; the remaining decoder and output head are retained.

Finally, the decoder progressively restores spatial resolution and combines the deep change representation with available skip features. A final prediction layer produces four logits per pixel, from which the background, intact, damaged, or destroyed label is selected. During backpropagation, the four-class label map supervises both ordinary and DA-CutMix-augmented samples. At inference, DA-CutMix is absent, whereas the inserted CMCA module remains part of the network. This division of labor is central to the proposed design: CMCA addresses how the two sensors interact inside the model, and DA-CutMix addresses which rare cross-event examples the model observes during training.

C. Modality-Specific Encoder

The six-channel tensor used by the common training interface is split back into the optical and SAR tensors before feature extraction. Let $l \in \{1,\dots,L\}$ index the encoder stages and let $m \in \{o,s\}$ identify the optical or SAR modality. With $\mathbf{F}_0^o = \mathbf{X}^o$ and $\mathbf{F}_0^s = \mathbf{X}^s$, the two feature hierarchies are

$$\mathbf{F}_l^m = E_l^m\left(\mathbf{F}_{l-1}^m\right), \tag{2}$$

where E_l^m is stage l of the encoder for modality m, and $\mathbf{F}_l^m \in \mathbb{R}^{C_l\times h_l\times w_l}$ is its output feature map. C_l is the number of feature channels at that stage. The lowercase h_l and w_l denote the height and width of an intermediate feature map; they are distinguished from the uppercase input dimensions H and W because spatial downsampling generally gives $h_l < H$ and $w_l < W$. The batch dimension is omitted from the mathematical notation for readability.

For an encoder–decoder backbone with skip connections, paired shallow features are combined as

$$\mathbf{S}_l = \eta_l([\mathbf{F}_l^o \| \mathbf{F}_l^s]), \tag{3}$$

where $\|$ denotes concatenation along the channel dimension, η_l is a learned 1×1 projection, and $\mathbf{S}_l$ is the fused skip feature supplied to the corresponding decoder stage. The projection reduces the concatenated $2C_l$ channels to the channel count expected by the original decoder. Backbones without explicit skip connections do not require (3); they only expose a paired deep feature map to CMCA. This common two-input/one-output interface is what allows the fusion module to be attached to different encoder families.

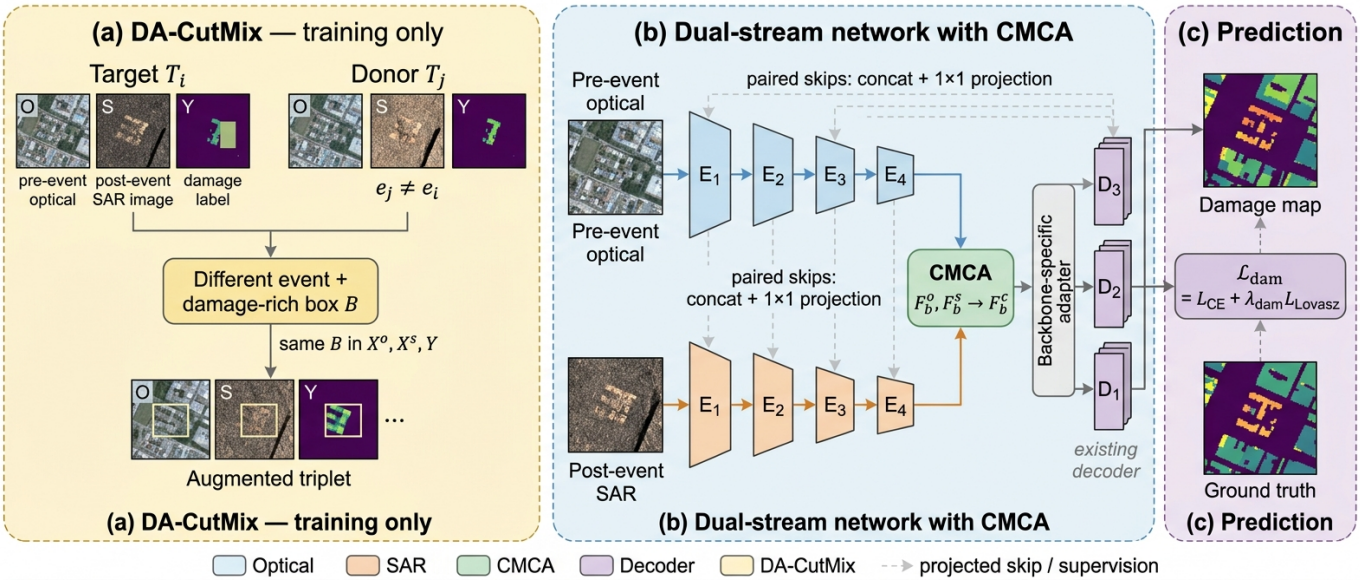


Fig. 1. Technical roadmap of the proposed multimodal building damage assessment framework. The pipeline contains architecture-level optical/SAR fusion through CMCA and data-level cross-event augmentation through DA-CutMix.

D. Cross-Modal Change Attention

Direct channel concatenation leaves the network to discover cross-sensor correspondences implicitly. CMCA instead makes the comparison directional: the pre-event optical representation asks where structurally relevant post-event evidence can be retrieved from the SAR representation. The module operates at a selected deep encoder stage, denoted by subscript b, because deep features are more semantic and have fewer spatial positions than shallow features. Its detailed architecture is shown in Fig. 2.

Let $\mathbf{F}_b^o, \mathbf{F}_b^s \in \mathbb{R}^{C_b \times h_b \times w_b}$ be the paired optical and SAR features at this stage. Here, C_b is their common channel dimension, h_b and w_b are their feature-map height and width, and $N = h_b w_b$ is the number of optical spatial positions. The batch dimension is again omitted. With M attention heads, each head has dimension $d = C_b/M$. The input features are projected and reshaped as

$$\begin{aligned} \mathbf{Q} &= \mathcal{R}_M(W_q \mathbf{F}_b^o)\,, \\ \mathbf{K} &= \mathcal{R}_M(W_k \mathcal{S}_r(\mathbf{F}_b^s))\,, \\ \mathbf{V} &= \mathcal{R}_M(W_v \mathcal{S}_r(\mathbf{F}_b^s))\,. \end{aligned} \tag{4}$$

W_q, W_k, and W_v are learned 1×1 convolutions. $\mathcal{R}_M$ reshapes a feature map into M head-specific token sequences, giving $\mathbf{Q} \in \mathbb{R}^{M\times N\times d}$ and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{M\times N_s\times d}$. $\mathcal{S}_r$ is a depth-wise convolution with kernel size and stride r that reduces only the SAR key and value maps. Consequently, $N_s = h_s w_s$ is the number of reduced SAR positions, with h_s and w_s denoting the reduced height and width.

For each head, scaled dot-product attention first measures the similarity between each optical query and every reduced SAR key:

$$\mathbf{A} = \mathrm{softmax}_{N_s}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right). \tag{5}$$

Here, $\mathbf{A} \in \mathbb{R}^{M\times N\times N_s}$ is the cross-modal attention map. $\mathbf{K}^\top$ is transposed over its last two dimensions, $\mathbf{Q}\mathbf{K}^\top$ contains query–key dot products, and division by $\sqrt{d}$ prevents their magnitude from growing with the head dimension. $\mathrm{softmax}_{N_s}$ normalizes each row over the N_s SAR locations, so every row of $\mathbf{A}$ is a set of nonnegative retrieval weights that sums to one. A larger weight means that the learned optical query and SAR key are more similar in the projected feature space.

The weights aggregate the SAR values and the heads are reshaped back to a feature map:

$$\tilde{\mathbf{F}}_b^s = \mathcal{R}_M^{-1}(\mathbf{A}\mathbf{V}), \tag{6}$$

where $\mathbf{A}\mathbf{V}$ is a weighted sum of SAR value vectors for every optical position, and $\mathcal{R}_M^{-1}$ merges the M heads and restores the two-dimensional layout. The resulting $\tilde{\mathbf{F}}_b^s \in \mathbb{R}^{C_b\times h_b\times w_b}$ is not a geometrically warped SAR image; it is an optical-conditioned SAR feature representation.

The retrieved SAR feature and original optical feature are compared by a learned projection:

$$\mathbf{F}_b^c = \phi\Big([\tilde{\mathbf{F}}_b^s \| \mathbf{F}_b^o]\Big)\,, \tag{7}$$

where $\mathbf{F}_b^c \in \mathbb{R}^{C_b\times h_b\times w_b}$ is the CMCA output and ϕ consists of a 1×1 convolution, batch normalization, and ReLU. Unlike a fixed subtraction, ϕ learns which combinations of pre-event structure and retrieved post-event response are useful for the final damage loss.

Plug-in connection to the base network: the core CMCA interface always maps two equally sized feature maps to one feature map of the same size. A small adapter then restores the interface expected by the unmodified downstream network. Three adapters are used in our experiments. For encoder–decoder models whose bottleneck expects the jointly fused modalities, the concatenation adapter is

$$\mathbf{F}_b^{\mathrm{out}} = \psi([\mathbf{F}_b^o \| \mathbf{F}_b^s \| \mathbf{F}_b^c])\,, \tag{8}$$

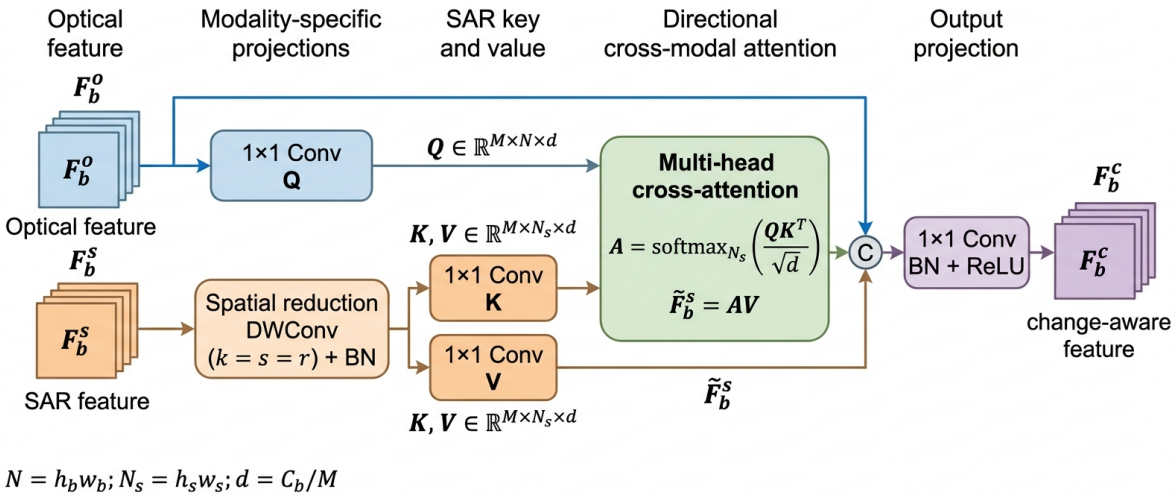


Fig. 2. Detailed architecture of the proposed Cross-Modal Change Attention module. Optical bottleneck features generate queries, while SAR bottleneck features generate keys and values after spatial reduction.

where ψ is a learned projection that maps the $3C_b$ concatenated channels to the channel count required by the original decoder. A replacement adapter uses $\mathbf{F}_b^{\text{out}} = \mathbf{F}_b^c$ when the decoder already expects C_b channels. A residual adapter uses $\mathbf{F}_b^{\text{out}} = \mathbf{F}_b^{\text{base}} + \rho(\mathbf{F}_b^c)$, where $\mathbf{F}_b^{\text{base}}$ is the backbone's original fused feature and ρ matches its channel dimension. These choices do not change the attention computation; they only connect its output to different backbone interfaces. The exact adapter and insertion stage for each evaluated backbone are listed in Table IV.

E. Damage-Aware CutMix

This module is designed specifically to increase the training exposure of the two scarce and operationally important classes: damaged buildings (class 2) and destroyed buildings (class 3). Background and intact-building pixels dominate BRIGHT, while the appearance and surroundings of severe damage vary across disasters. DA-CutMix therefore extends CutMix [18] with two constraints: the donor must come from a different disaster event, and the accepted donor box must contain enough class-2 or class-3 pixels. Its purpose is not to synthesize a physically complete disaster scene. Instead, it presents rare damage evidence under a different event context while retaining the local optical–SAR–label correspondence inside the pasted region.

Each training sample is represented as

$$\mathcal{T}_i = (\mathbf{X}_i^o, \mathbf{X}_i^s, \mathbf{Y}_i, e_i)\,, \tag{9}$$

where i is the index of the target sample in the training list, $\mathbf{X}_i^o$ and $\mathbf{X}_i^s$ are its optical and SAR images, $\mathbf{Y}_i$ is its four-class label map, and e_i is its disaster-event identifier parsed from the file name.

Before mixing, a single set of random geometric parameters is applied to all three spatially aligned elements of $\mathcal{T}_i$. The optical image, SAR image, and label map receive the same crop coordinates, horizontal or vertical flip, and rotation. Thus, synchronized augmentation cannot move one modality without moving the other modality and its labels in exactly the same way.

For every augmented target $\mathcal{T}_i$, a Bernoulli trial with probability p decides whether DA-CutMix is attempted. Here, $p \in [0,1]$ is the application probability: $p = 0$ always retains the target, while $p = 1$ always attempts mixing. If the trial fails, the ordinary augmented target is returned. If it succeeds, a donor index j is sampled such that $e_j \neq e_i$. The donor triplet $\mathcal{T}_j$ receives its own synchronized geometric augmentation, and a rectangular box B is sampled from the donor label. The box is accepted only when

$$\sum_{\mathbf{u}\in B} \mathbb{I}\left[\mathbf{Y}_j(\mathbf{u}) \in \mathcal{C}_d\right] \geq \max\left(n_{\min}, \tau|B|\right), \tag{10}$$

where $\mathbf{u}$ is a pixel coordinate inside B, $\mathbb{I}[\cdot]$ is 1 when its enclosed condition is true and 0 otherwise, and $\mathcal{C}_d = \{2,3\}$ is the targeted set containing damaged and destroyed buildings. The left-hand side therefore counts targeted damage pixels in the candidate box. $|B|$ is the total number of pixels in the box, $n_{\min}$ is the required absolute number of targeted pixels, and $\tau \in [0,1]$ is the required targeted-pixel fraction. Taking the maximum enforces both an absolute and a relative requirement. Rejection sampling does not require every pixel in B to be damaged; background and intact pixels may remain to preserve local context around the damaged structures.

Let $\mathbf{M}_B \in \{0,1\}^{H\times W}$ be a binary mask that equals 1 inside the accepted box and 0 elsewhere. The mixed optical image, SAR image, and label map are generated jointly:

$$\begin{aligned}
\tilde{\mathbf{X}}^o &= \mathbf{M}_B \odot \mathbf{X}_j^o + (1 - \mathbf{M}_B) \odot \mathbf{X}_i^o, \\
\tilde{\mathbf{X}}^s &= \mathbf{M}_B \odot \mathbf{X}_j^s + (1 - \mathbf{M}_B) \odot \mathbf{X}_i^s, \\
\tilde{\mathbf{Y}} &= \mathbf{M}_B \odot \mathbf{Y}_j + (1 - \mathbf{M}_B) \odot \mathbf{Y}_i.
\end{aligned} \tag{11}$$

In (11), $\odot$ denotes element-wise multiplication, $1 - \mathbf{M}_B$ selects the area outside the box, subscript j identifies donor content, subscript i identifies retained target content, and the tilde denotes the resulting mixed sample. The two-dimensional mask is broadcast across the three image channels. Using the

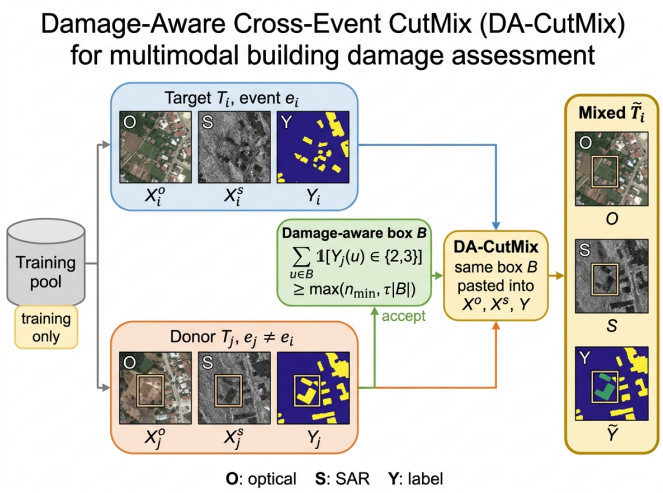


Fig. 3. Training pipeline of DA-CutMix. Donor patches are sampled from different disaster events and must contain a minimum amount of damaged or destroyed pixels before the same spatial box is pasted jointly into the optical image, SAR image, and label map.

identical mask in all three lines preserves the pre-event optical, post-event SAR, and label alignment within the donor region. If no box satisfies (10) after the configured candidate and donor trials, the original augmented target is used unchanged. Because DA-CutMix returns a triplet with exactly the same shapes and data types as its input, it can be enabled in the data loader without modifying any backbone. It is disabled during validation and inference and therefore introduces no test-time operation.

F. Training Objective

The damage-classification branch is optimized using multi-class cross-entropy and Lovasz-Softmax losses [32]:

$$\mathcal{L}_{\text{dam}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{dam}} \mathcal{L}_{\text{Lovasz}}. \tag{12}$$

Here, $\mathcal{L}_{\text{CE}}$ provides pixel-wise semantic supervision, whereas $\mathcal{L}_{\text{Lovasz}}$ directly optimizes a differentiable surrogate of class-wise intersection-over-union (IoU), thereby reducing the influence of class imbalance on region overlap. The coefficient λ_{dam} controls the contribution of the Lovasz-Softmax term.

Two output configurations are considered. A single-head network directly predicts the four-class damage map and is optimized using (12). A dual-head network additionally predicts a binary building-localization map as an auxiliary task. For the latter configuration, the total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{dam}} + \mathcal{L}_{\text{CE}}^{\text{loc}} + \lambda_{\text{loc}} \mathcal{L}_{\text{Lovasz}}^{\text{loc}}, \tag{13}$$

where the superscript loc denotes supervision of the auxiliary building-localization branch and λ_{loc} weights its Lovasz-Softmax term. This branch distinguishes building pixels from the background and provides additional spatial supervision for damage classification. The choice between (12) and (13) depends only on the output structure of the base network; neither CMCA nor DA-CutMix introduces an additional loss term.

TABLE I
DISASTER TYPES AND EVENT LOCATIONS IN BRIGHT.

Disaster type	Event location(s)
Earthquake	Haiti, Morocco, Noto, and Turkey
Flood	Libya
Hurricane	Mexico and Myanmar
Volcano	Congo and La Palma
Wildfire	Hawaii and Marshall
Explosion	Bata and Beirut
Conflict	Ukraine

TABLE II
DAMAGE LABEL DEFINITIONS IN THE BRIGHT DATASET.

Category	Definition
Background (0)	Non-building pixels.
Intact (1)	Building pixels without visible structural damage or apparent disaster-related surface changes.
Damaged (2)	Buildings with partial structural damage, such as missing roof components, visible cracks, partial wall or roof collapse, or surrounding hazard effects from water, mud, burning, or volcanic material.
Destroyed (3)	Buildings that are completely collapsed, burned, severely covered by water or mud, or no longer visible at the annotated location.

IV. EXPERIMENTS

A. Dataset and Experimental Protocol

Experiments are conducted on the BRIGHT multimodal building damage assessment benchmark [10]. BRIGHT provides coregistered pre-event optical images, post-event SAR images, and pixel-level building damage annotations at very high spatial resolution. It covers 14 disaster events in 23 regions. The events span five natural-disaster types—earthquake, flood, hurricane, volcanic eruption, and wildfire—and two human-made types—explosion and armed conflict. Table I lists the locations used to identify these events in the standard data split.

The annotations contain four semantic categories: background, intact building, damaged building, and destroyed building, as summarized in Table II. Following the standard supervised split provided with BRIGHT, 2,917 patch records are used for training, 449 for validation, and 877 for testing. The validation set is used for model selection, and final performance is reported on the test set.

B. Evaluation Metrics

We report overall accuracy (OA), per-class intersection over union (IoU), mean IoU (mIoU), building-localization F1 ($F1_{\text{loc}}$), and damage-classification F1 ($F1_{\text{clf}}$). Let $K = 4$ be the number of semantic classes and N the total number of evaluated pixels. For class c, TP_c, FP_c, and FN_c denote the numbers of true-positive, false-positive, and false-negative pixels, respectively.

OA measures the proportion of correctly classified pixels over the complete damage map:

$$\text{OA} = \frac{\sum_{c=0}^{K-1} \text{TP}_c}{N}. \tag{14}$$



Fig. 4. Example DA-CutMix training sample from the BRIGHT dataset. From left to right: pre-event optical image, post-event SAR image, and damage label map. The yellow rectangle denotes the same cross-event donor region pasted into all three layers, illustrating that DA-CutMix preserves optical–SAR–label alignment while injecting a damage-rich patch.

A larger OA indicates higher overall pixel accuracy, although this metric can be dominated by the abundant background class.

The overlap between the prediction and ground truth for class c is measured by

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}. \tag{15}$$

The mean over the four semantic classes is

$$\mathrm{mIoU} = \frac{1}{K} \sum_{c=0}^{K-1} \mathrm{IoU}_c. \tag{16}$$

Higher IoU indicates better spatial agreement for the corresponding class, while higher mIoU indicates better class-balanced segmentation performance. We report the four class-specific IoUs to distinguish background, intact, damaged, and destroyed performance.

For building localization, labels 1–3 are merged into a binary building foreground, irrespective of damage severity. Let TP_b, FP_b, and FN_b be the resulting binary pixel counts. The localization score is

$$F1_\mathrm{loc} = \frac{2\mathrm{TP}_\mathrm{b}}{2\mathrm{TP}_\mathrm{b} + \mathrm{FP}_\mathrm{b} + \mathrm{FN}_\mathrm{b}}. \tag{17}$$

A larger $F1_\mathrm{loc}$ indicates more accurate building delineation without considering whether the damage level is correct.

Damage classification is evaluated on ground-truth building pixels. For each building class $c \in \{1, 2, 3\}$, its F1 score is

$$F1_c = \frac{2\mathrm{TP}_c}{2\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad c \in \{1, 2, 3\}, \tag{18}$$

where the three classes correspond to intact, damaged, and destroyed buildings. Their harmonic mean gives

$$F1_\mathrm{clf} = \frac{3}{\sum_{c=1}^{3} F1_c^{-1}}. \tag{19}$$

A larger $F1_\mathrm{clf}$ indicates more balanced damage recognition; because it is a harmonic mean, poor performance on any building class reduces the final score. All metrics range from 0 to 1 and are reported as percentages. Higher values are better, and 100% denotes perfect agreement under the corresponding definition.

TABLE III
TRAINING AND DA-CUTMIX HYPERPARAMETERS.

Parameter	Setting
Training crop size	640×640 pixels
Training batch size	16
Training iterations	50,000
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	5×10^{-3}
Loss weights	$\lambda_\mathrm{dam} = 0.75$; $\lambda_\mathrm{loc} = 0.5$
DA-CutMix target classes	$\mathcal{C}_d = \{2, 3\}$
DA-CutMix probability	$p \in \{0.25, 0.30\}$
Minimum targeted pixels	$n_\mathrm{min} = 200$
Minimum targeted ratio	$\tau = 0.05$
Patch height/width ratio	Independently sampled from $[0.12, 0.35]$

C. Implementation Details

All methods use the same data split and optimization protocol, with AdamW [33] as the optimizer. The principal training settings and DA-CutMix hyperparameters are summarized in Table III.

The CMCA core is unchanged across backbones, while its channel dimension, head count, spatial-reduction factor, and output adapter are matched to the selected insertion feature, as summarized in Table IV.

Random cropping, horizontal and vertical flipping, and rotations by multiples of 90° are applied synchronously to the optical image, SAR image, and label map. The two images are normalized separately. DA-CutMix is applied after geometric augmentation and before normalization. At most 10 donor samples and 10 candidate boxes per donor are tested before the original target is returned unchanged. Its application probability p is selected from $\{0.25, 0.30\}$ on the validation set.

TABLE IV
BACKBONE-SPECIFIC CMCA INSERTION AND ATTENTION CONFIGURATION FOR A 640×640 CROP.

Backbone	Paired insertion feature	Output adapter	C_b	$h_b \times w_b$	Heads M	Reduction r
UNet	Deepest convolutional features	Concatenate–project	512	40×40	4	2
SiamAttnUNet	Fifth encoder-stage features	Replacement	1024	40×40	8	2
DamageFormer	Fused Swin pyramid features	Concatenate–project	256	160×160	4	16
SiamCRNN	ResNet-18 layer-4 features	Residual injection	512	20×20	8	1
DeepLabV3Plus	ResNet-50 high-level features	Concatenate–project	2048	40×40	8	2
ChangeMamba	Deepest VSSM features	Residual injection	768	20×20	8	1

D. Compared Variants

We evaluate the two proposed components under six representative segmentation backbones: UNet [34]; DamageFormer, the benchmark adaptation of DamFormer [26] with Swin-T encoders [35]; SiamAttnUNet [8]; SiamCRNN [24]; DeepLabV3Plus [36]; and ChangeMamba [27]. UNet, SiamAttnUNet, and DeepLabV3Plus use the single-head objective in (12), whereas DamageFormer, SiamCRNN, and ChangeMamba retain the auxiliary localization head and use the dual-head objective in (13). For each backbone, the baseline denotes the original model trained without CMCA or DA-CutMix. The "+CMCA" setting exposes the paired features listed in Table IV, applies the same CMCA computation, and reconnects the result through the corresponding adapter while retaining the downstream decoder or head. The "+DA-CutMix" setting leaves the architecture unchanged and enables the proposed cross-event, damage-aware transform in the training loader only. The "+CMCA+DA-CutMix" setting combines these independently switchable model- and data-level components. When a compact final comparison reports "+CMCA+DA-CutMix" without an explicit probability, it denotes the validation-selected CutMix probability rather than a single probability shared by all backbones.

E. Ablation Study

Table V reports the ablation results on the standard test split for the baseline, CMCA, DA-CutMix, and their combination across six backbones. For configurations involving DA-CutMix, the probability p is selected from $\{0.25, 0.30\}$ using the validation set and omitted from the method names for clarity. The combined configuration achieves the highest mIoU for five backbones, improving their corresponding baselines by 0.39–1.94 percentage points. This consistency across different architectures indicates that model-level cross-modal interaction and data-level damage-aware augmentation are generally complementary.

Among the convolutional and Siamese backbones, the combined configuration improves mIoU from 64.74% to 66.16% for UNet, from 65.03% to 66.28% for SiamAttnUNet, from 66.13% to 66.52% for SiamCRNN, and from 63.85% to 65.79% for DeepLabV3Plus. The largest gain is obtained with DeepLabV3Plus. These results indicate that explicit optical–SAR feature interaction is particularly useful for architectures without dedicated heterogeneous comparison, while DA-CutMix further improves their exposure to scarce damage classes.

The two stronger backbones exhibit different behaviors. For DamageFormer, standalone DA-CutMix produces the best mIoU, increasing it from 66.84% to 67.20%, whereas the combined configuration reaches 67.14%. In contrast, ChangeMamba benefits most from the combined configuration, which improves mIoU from 66.73% to 67.43%. These results suggest that the relative contribution of the two modules is backbone-dependent: damage-aware sampling contributes more strongly to DamageFormer, whereas explicit optical–SAR interaction remains complementary to the long-range context modeling of ChangeMamba.

F. Per-Event Robustness

Table VI compares the reported configuration of each backbone with its baseline using the unweighted mean over 14 events. The listed configurations improve the event average for five backbones and outperform their baselines on 7–12 events. UNet and DeepLabV3Plus show the largest average gains of 3.50 and 2.29 percentage points, respectively, indicating that their aggregate improvements are distributed across multiple disaster scenarios rather than being driven by a single event.

DamageFormer is the only exception: standalone DA-CutMix with $p = 0.30$ gives its best global mIoU but reduces the event average by 0.36 percentage points. This difference between global and event-level performance shows that the contribution of damage-aware mixing is backbone-dependent and motivates reporting both evaluation views.

TABLE V
OVERALL ABLATION RESULTS ON THE BRIGHT BENCHMARK. VALUES ARE REPORTED IN PERCENTAGE.

Method	$F1_{loc}$	$F1_{clf}$	OA	mIoU	Background	Intact	Damaged	Destroyed
UNet	88.32	70.91	95.52	64.74	96.28	71.62	37.48	53.59
UNet + DA-CutMix	88.07	70.89	95.48	64.46	96.22	71.43	38.05	52.15
UNet + CMCA	88.68	**72.65**	95.59	65.39	96.29	72.79	**39.30**	53.20
UNet + CMCA + DA-CutMix	**89.18**	72.43	**95.81**	**66.16**	**96.50**	**73.58**	39.08	**55.49**
DamageFormer	89.88	72.61	96.00	66.84	96.70	74.62	39.97	56.06
DamageFormer + DA-CutMix	**89.94**	73.05	**96.09**	**67.20**	**96.77**	**74.94**	**41.40**	55.70
DamageFormer + CMCA	89.90	71.96	95.97	66.55	96.72	74.35	38.61	56.53
DamageFormer + CMCA + DA-CutMix	89.91	**73.07**	96.00	67.14	96.71	74.54	40.17	**57.12**
SiamAttnUNet	88.26	71.79	95.51	65.03	96.21	72.06	38.24	53.60
SiamAttnUNet + DA-CutMix	87.92	71.85	95.49	65.28	96.11	71.95	37.45	55.62
SiamAttnUNet + CMCA	88.08	72.15	95.49	65.05	96.19	71.67	**39.50**	52.84
SiamAttnUNet + CMCA + DA-CutMix	**88.44**	**72.67**	**95.67**	**66.28**	**96.33**	**72.55**	38.94	**57.30**
SiamCRNN	**89.11**	72.12	95.83	66.13	**96.49**	73.63	40.22	54.17
SiamCRNN + DA-CutMix	89.05	72.15	**95.85**	66.36	**96.49**	**73.67**	39.20	**56.09**
SiamCRNN + CMCA	88.85	72.85	95.75	66.25	96.42	73.12	**41.26**	54.21
SiamCRNN + CMCA + DA-CutMix	89.04	**73.13**	95.75	**66.52**	96.41	73.49	40.10	56.06
DeepLabV3Plus	87.36	69.91	95.39	63.85	96.05	70.82	37.06	51.45
DeepLabV3Plus + DA-CutMix	87.68	**71.77**	95.53	65.25	96.10	71.81	**39.80**	53.30
DeepLabV3Plus + CMCA	87.71	69.76	95.56	64.83	96.23	71.14	38.18	53.76
DeepLabV3Plus + CMCA + DA-CutMix	**88.79**	71.22	**95.80**	**65.79**	**96.44**	**73.32**	39.26	**54.12**
ChangeMamba	89.95	73.18	95.83	66.73	96.42	73.66	40.74	56.09
ChangeMamba + DA-CutMix	89.75	73.54	95.89	67.04	96.47	73.88	42.21	55.60
ChangeMamba + CMCA	89.93	**74.41**	95.87	67.32	96.45	73.95	**42.55**	56.34
ChangeMamba + CMCA + DA-CutMix	**90.12**	74.31	**95.92**	**67.43**	**96.54**	**74.12**	41.82	**57.25**

TABLE VI
PER-EVENT MIOU OF THE REPORTED CONFIGURATIONS ON BRIGHT. VALUES ARE REPORTED IN PERCENTAGE.

Backbone	Reported configuration	Baseline avg.	Config. avg.	Δ avg.	Improved events
UNet	+CMCA+DA-CutMix	48.09	51.59	$+3.50$	12/14
DamageFormer	+DA-CutMix	52.04	51.68	-0.36	7/14
SiamAttnUNet	+CMCA+DA-CutMix	48.49	49.85	$+1.36$	11/14
SiamCRNN	+CMCA+DA-CutMix	50.29	51.11	$+0.82$	7/14
DeepLabV3Plus	+CMCA+DA-CutMix	46.79	49.08	$+2.29$	12/14
ChangeMamba	+CMCA+DA-CutMix	50.39	51.82	$+1.43$	10/14

V. CONCLUSION

In this article, we propose two independently pluggable components, CMCA and DA-CutMix, for multimodal building damage assessment using pre-event optical and post-event SAR imagery. CMCA is inserted at a deep encoder stage to model cross-modal relationships by using optical features to retrieve relevant SAR responses, thereby enhancing the representation of disaster-induced changes. DA-CutMix is applied during training to paste damage-rich regions from different disaster events synchronously into the optical image, SAR image, and label map. In this way, the network observes more damaged and destroyed samples while the spatial correspondence between the two modalities and their labels is preserved. Both components can be integrated into different segmentation backbones without redesigning the original decoder, and DA-CutMix introduces no additional inference-time computation. Experimental results on the BRIGHT benchmark with six representative backbones demonstrate the effectiveness of the proposed components. The combined configuration achieves the highest mIoU within five of the six backbone groups, while the overall and per-event results show that the relative contributions of CMCA and DA-CutMix depend on the backbone and disaster distribution.

Future work will focus on three directions. First, an adaptive DA-CutMix strategy will be developed to determine the donor event, target class, patch scale, and mixing probability according to class occurrence and training difficulty. Second, cross-event generalization will be further evaluated and improved under leave-one-event-out, leave-one-disaster-type-out, and cross-dataset settings. Third, multiscale and SAR-specific scattering information will be incorporated into CMCA, while its computational cost will be further reduced for VHR imagery.

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